

When Is an LLM Forecasting Benchmark Actually Forecasting? Source Currency, Equal Information, and Harnessing Forecast Signal

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Abstract

Recent work increasingly evaluates LLM forecasters against human crowds, prediction markets, and live trading outcomes. Such comparisons are only meaningful when all sides are measured under the same information state. This paper asks whether LLM forecast signal can be harnessed once source currency, label-time validity, and equal-information baselines are enforced. We study five model families with more than 20,000 persisted calls in a SQLite database. The central validity and market-control claims use public prediction-market and official-data rows; a private low-overlap corpus contributes secondary elicitation diagnostics only. The foundation is a source-currency effect: in a matched Manifold panel of 80 contracts and 240 tool-free model calls, post-cutoff rows are harder than pre-cutoff/source-visible rows by +0.191098 Brier in aggregate, with paired-stratum delta +0.2155 and permutation $p = 0.0004$. The result survives a partial pre-outcome market-prior repair on 51 contracts and an adversarial missing-band sensitivity check. The same controls do not support a raw LLM-superiority claim: the database now has 103 external market baseline rows and 52 equal-information market rows. On a 24-contract Polymarket replacement slice, Claude, Codex, Gemini, and DeepSeek all have higher Brier than the market baseline: the four-family mean-probability panel Brier is 0.267758, Polymarket Brier is 0.072964, and paired panel-minus-market $p = 0.0068$. On a separate 24-contract Manifold fill, the market also scores better, but the comparison is inconclusive: five-family low-stake model-panel Brier 0.198723, Manifold Brier 0.160977, paired $p = 0.5431$. The constructive result is a controlled way to use the remaining model signal. A source-valid low-probability calibration rule improves every tested family on the public-domain panel and leaves the raw mean-panel +0.029598 Brier worse on source- and label-time-documented rows. A source-balanced pairwise-ranking packet reaches 0.750 accuracy over 24 non-tie pairs. Structured evidence fields and family-choice headroom identify additional signal, although they do not yet supply a reliable automated selection rule. In contrast, simple prompt intervention and self-repair do not reliably improve forecasts.

1 Introduction

This paper asks two questions: when is an LLM forecasting benchmark actually measuring future-event prediction, and what has to be done before the resulting model signal is usable?

The central result is a controlled-use claim. Some model signal remains useful after invalid comparisons are removed, and that usefulness depends on the information state of the row. A benchmark row should not be used for broad conclusions until three conditions are documented. First, *source currency*: the resolved answer was not already source-visible to the model generation being tested. Second, *label-time validity*: the outcome label matches what would have been knowable under the forecast-time data vintage. Third, *equal-information baselines*: human or market comparison bars are measured on the same contract under the same pre-outcome information rule. In our data,

the source-currency result is strong enough to report as a measurement result. Broad human- or market-superiority claims are not: the database has 103 typed external market baseline rows and 52 ingested equal-information market rows; Polymarket beats the four-family model panel on the same 24 contracts, and the separate Manifold fill favors the market but is inconclusive.

The empirical order matters. A black-box LLM forecaster first has to be tested on rows that are valid for its generation. Only then does it make sense to ask which emitted channel is useful for a particular family and source: point estimate, worry scalar, bid-ask spread, self-predicted Brier interval, reference-class base rate, or cross-family disagreement. Only after those two questions can we ask whether a prompt, abstention rule, review policy, or allocation rule improves Brier or utility against explicit controls.

The paper therefore has one main argument with two parts. The validity part is that source-currency, label-time, and equal-information checks change what one is allowed to conclude from LLM forecasting benchmarks. The constructive part is that the surviving model signal is specific rather than global: overconfident low-probability forecasts can be corrected on source-valid forward-looking rows; pairwise comparisons can rank harder/easier contracts better than chance under source-balanced controls; and family-choice differences create measurable headroom that current simple selection rules do not yet recover. The bias-transfer material is kept only where it explains why prompt-only interventions were fragile. This separation prevents the manuscript from becoming an inventory of experiments.

Contributions. The paper makes four contributions. First, it states a documentation test for LLM forecasting rows: a row is not broad forecast evidence unless source currency, label-time validity, and equal-information comparison status are known. Second, it gives an empirical source-currency audit in which post-cutoff Manifold rows are substantially harder than matched pre-cutoff/source-visible rows. Third, it reports two same-information market controls that show why raw LLM panels are not the right unit of comparison. Fourth, it identifies the model-derived signals that still work under controls: a source-valid low-probability calibration rule, source-balanced pairwise ranking, mixed but informative structured evidence-field tests, and family-choice headroom. These are positive results, but they are not a market- or human-superiority claim.

Positioning. The closest related work now falls into four groups: future-question benchmarks such as ForecastBench [8], system papers such as AIA Forecaster [1], belief-updating benchmarks such as EVOLVECAST [15], and market or replay environments such as Prediction Arena, PolyBench, PredictionMarketBench, Foresight Arena, MarketBench, and Reppo-style market infrastructure [16, 3, 2, 10, 4, 13]. Those systems ask how well agents forecast, update, trade, or coordinate through market-like mechanisms. This paper asks what has to be true before such comparisons are interpretable at the row level. It is therefore closest in spirit to recent evaluation-warning work on temporal leakage and benchmark extrapolation [11], but adds a scored empirical audit and controlled mechanisms for using model signal after the validity checks.

Evidence discipline. Table 1 states the paper’s main claims and their limits. It is included near the front because the main risk in this area is not a missing benchmark but a wrong comparison: rows that are source-visible, labels that use a later data vintage, market bars measured at a different information time, or decision rules adopted from diagnostic correlations.

The paper is empirical first and theoretical second. We ran a calibration study across five model families and two corpus classes, with more than 30 measured findings and pre-registered tests. The main text keeps the forecasting spine in front: validity checks, same-information baselines, and

Claim unit	Evidence in this paper	Conservative interpretation
Validity checks	Source-currency, label-time, and equal-information audits over the database	A row without these checks is not broad forecast evidence.
Raw LLM panels vs markets	Polymarket scores much better than the model panel; Manifold also scores better, but inconclusively	No broad market/human superiority claim.
Low-probability calibration	Source-valid low-probability rule improves every family on the public-domain panel and improves the forward-looking slice	Point-probability use for source-valid rows, not retrospective correction.
Pairwise ranking	Source-balanced pairwise ranking and partial probability-translation evidence	Ranking/tournament support; absolute-probability use needs larger controls.
Evidence fields and family choice	Mixed structured-field tests; best-family-in-hindsight headroom not recovered by simple allocation rules	Evidence of where signal lives; selection rules need prospective confirmation.
Prompt intervention/self-repair	Selective action, self-repair, and diagnostic allocation mostly fail controls	Evidence against unvalidated prompt-only interventions.

Table 1: Main claims and limits. The central contribution is the validity layer plus the controlled mechanisms that can extract or locate LLM forecast signal once invalid comparisons are removed.

Stage	Question answered	Consequence
Forecast row	What probability, source, timestamp, and side channels did the model emit?	Defines the unit to be scored.
Validity checks	Was the answer source-visible, and is the label valid at the relevant data vintage?	Decides whether the row can count as forecast evidence.
Same-information baseline	What did a market or human comparator know at the same pre-outcome time?	Decides whether raw model probabilities add value under the same information rule.
Controlled use	Which model-derived signal improves a proper score or ranking task after the checks above?	Admits only source-valid calibration, pairwise ranking, or future family-choice rules that beat controls.

Figure 1: Evidence flow. The paper’s unit is a forecast row with documented validity checks, a same-information baseline, and a controlled decision about which model-derived signal is usable.

controlled uses of model signal. Bias-transfer and prompt-stability results are treated as secondary diagnostics because they help explain why generic prompting is unreliable, but they do not carry the paper’s main claim.

Statistical audit. Every empirical claim is checked with the same power-aware procedure: Fisher- z sample-size computation before an experiment runs, three possible outcomes after scoring (supported, ruled out at the target effect size, or underpowered), equivalence testing instead of treating “ $p > 0.05$ ” as evidence of no effect, BH-FDR across panel tests, and leave-one-out R^2 at small N . Under this audit, 8 of 12 prior nulls in the study became underpowered rather than negative findings, and 5 cross-corpus claims narrowed to corpus-specific results. That statistical procedure is a methodological contribution because the same overcalling problem plausibly affects existing $N=5\text{--}20$ cross-corpus headlines. A second-order application is corpus-validity drift: a benchmark whose resolution dates were post-cutoff for publication-era LLMs can become pre-cutoff for the next model generation. We apply this point to the Halawi 2024 dataset [5] in §10.

Reading order. Section §2 defines the panel, corpora, and verdict rules; §3 states the paper’s core empirical results before the diagnostics. Sections §4–§5 give the elicitation-surface and family-heterogeneity measurements that motivated the validity audits. Section §7 reports the applied calibration and allocation consequences, including the source-valid low-probability calibration and pairwise-ranking results. Section §8 synthesizes the supported harnessing thesis and explains why this remains one integrated paper rather than a split paper. Section §9 gives the secondary bias-transfer and prompt-stability diagnostics. The main measurement audit is collected in §10: corpus-validity drift, source-currency checks, label-time checks, market-prior repair, and equal-information market baselines. The paper closes with explicit limits, next tests, a conclusion, and a reproducibility plan.

2 Setup

2.1 Five LLM families

Claude Opus 4.7 (Anthropic, via Claude Code CLI), GPT-5.5 and GPT-5.4-mini (OpenAI, via Codex CLI), Gemini 2.5 Flash (Google GenAI API), DeepSeek Chat (DeepSeek API). All API calls temperature=0 unless explicitly noted. All paths were audited for web-search-tool access; all prompts included an explicit instruction not to use web search.

2.2 Two corpora, four confounded axes

We use a *low-overlap corpus* of $N=15$ bespoke questions drawn from a private research domain with effectively zero LLM-pretraining overlap (formal-proof tactic combinations, state-transition questions from a private research workflow, and ground-truth-confirmed forecasting contracts), and a *public-domain corpus* of $N=42$ contracts from prediction markets (Manifold, Polymarket), stock-close thresholds, and ETF moves.

The four axes are confounded in the current data. Low-overlap-corpus questions are long, complex, unfamiliar, and without public priors *together*; public-domain questions are short, simple, familiar, and public-prior-rich *together*. The cross-corpus result (§5) is therefore evidence that *at least one* of these axes matters; a 4-cell de-confounded corpus design that breaks the confound is the highest-priority methodological follow-up (§11).

The low-overlap corpus is private and not publicly releasable in raw form; the reproducibility plan (§13) addresses this through a sanitized release with neutral identifiers and a parallel public-

Axis	Low-overlap ($N=15$)	Public-domain ($N=42$)
LLM pre-training overlap	Effectively zero	Substantial
Question length	247 chars (35 words)	99 chars (17 words)
Question structure	Multi-conditional, code/math symbols, parenthetical embedding	Atomic, simple syntactic structure
Implicit base rate	Ill-defined (bespoke operations)	Public-market priors well-trained

but-niche-academic corpus matching the four-axis profile. It is not required for the paper’s central source-currency, label-time, equal-information, market-control, or source-valid calibration claims. Those claims are scored from the public database, market-history packets, official-data label-time checks, and audit scripts listed in §13. Low-overlap rows are retained as secondary diagnostics about elicitation channels, prompt stability, and external generality.

2.3 Pre-registration and power-aware verdicts

Every experiment is pre-registered before launch with hypothesis, null, target effect size, n_{required} (computed via Fisher- z at $\alpha=0.05$ two-tailed, 80% power, Spearman correction +6%), falsifiers, and success criterion. Resolution uses three possible outcomes:

- **Supported:** 95% CI on observed ρ excludes 0.
- **Ruled out at the target effect size:** 95% CI wholly within $(-\text{target}, +\text{target})$; data rules out a meaningful effect.
- **Underpowered:** observed $|\rho|$ below detectability at the run N ; CI wide.

Brier-delta tests use paired-contract sign-flip permutation with a 90% bootstrap CI plus a BIC-approximation Bayes factor. “No effect” claims also require TOST equivalence at a pre-stated bound. Power thresholds at $\alpha=0.05$ / power= 0.80: $N=15 \Rightarrow |\rho| \geq 0.68$; $N=42 \Rightarrow 0.43$; $N=91 \Rightarrow 0.30$.

3 Core empirical results

The rest of the manuscript gives the diagnostics behind these claims. The core evidence is the following.

This ordering is deliberate. The paper’s constructive claims are downstream of the same-information market controls. A harnessing mechanism is included only when it either improves a proper score on valid rows, supports a controlled ranking use, or locates signal that a future decision rule could try to recover. Mechanisms that only change rationales, increase stated effort, or improve a source-visible retrospective slice are reported as diagnostics or failed interventions.

4 Why emitted forecast channels differ by family

Before reporting the elicitation findings, we name the structural distinctions that organized the measurements. These were not stated in advance; they are a compact reading of the body of findings the study produced. Their role in this paper is explanatory: they help explain why side channels and prompt interventions are family- and source-conditional, not why LLMs should be expected to beat markets.

Result	Evidence	Paper use
Source-currency validity matters	Matched Manifold Stage-B panel: 80 contracts / 240 tool-free calls; post-minus-pre Brier +0.191098, paired-stratum delta +0.2155, permutation $p = 0.0004$.	Candidate measurement result: forecast rows need source-currency checks.
Raw panels do not beat same-information markets	Polymarket replacement: panel 0.267758 vs market 0.072964, $p = 0.0068$. Manifold second source: panel 0.198723 vs market 0.160977, $p = 0.5431$.	Boundary: no LLM market/human superiority claim.
Low-probability calibration extracts point-probability signal under scope	Raw mean-panel remains +0.029598 Brier worse than the low-probability correction on rows with adequate source and label-time documentation, but the rule regresses source-visible rows.	Source-valid calibration view, not universal correction.
Pairwise ranking extracts relative signal under scope	Source-balanced pairwise packet: 24 unique non-tie pairs, accuracy 0.750, utility +0.583, $p = 0.0044$ vs random.	Ranking/tournament support, not standalone probability.
Prompt intervention/self-repair mostly fails	Selective action, generic self-repair, and diagnostic allocation fail or regress against controls.	Simple prompting is not enough.

Table 2: Core empirical results. The diagnostic sections explain how these results were found and where they fail; this table states the claims that remain after source, label-time, and market-baseline controls.

Same-information slice	Model panel Brier	Market Brier
Polymarket, 24 contracts	0.267758	0.072964
Manifold, 24 contracts	0.198723	0.160977

Figure 2: Equal-information market controls. Lower Brier is better. Polymarket clearly scores better than the model panel; Manifold also scores better, but inconclusively. These controls rule out a raw LLM-superiority claim for the current evidence.

Elicitation surface. An LLM forecaster, when asked, emits a set of channels: at minimum a point estimate p_{success} ; optionally a tail-worry scalar (“how worried are you?”), a bid-ask spread ($p_{\text{buy yes max}}, p_{\text{sell yes min}}$), a self-predicted Brier interval ($b_{\text{lo}}, b_{\text{hi}}$), an asymmetric-loss decision threshold under specified false-positive vs. false-negative cost regimes, or an outside-view base rate from analogous prior contracts. Training-mix differences across model families produce different elicitation surfaces, even when prompted identically. §5 introduces the bid-ask spread as a diagnostic error-warning channel that is elicitable on the low-overlap corpus; §5 shows the three uncertainty channels are statistically independent on a paired panel, motivating per-family channel analysis; §5 shows only one family generalises a multi-channel decomposition under leave-one-out cross-validation.

Behavioral structured-field interpretation. We do not observe the hidden activations of the closed models used here, so our channel claims are behavioral rather than mechanistic-interpretability claims. Still, the right unit is closer to a structured intermediate representation than to the generated rationale. Recent latent-prediction theory shows that learning or predicting hidden representations can avoid the sample-complexity cost of token-level prediction on hierarchical data [9]. Our setting is not training-time latent prediction, but the analogy is useful here: uncertainty channels and source-bound evidence fields can be scored against outcomes, while free-form rationale text is a noisy surface output. In two small follow-up tests across two families, structured evidence fields beat free prose on mean Brier, while the stricter two-step variant did not consistently beat a same-turn field. The combined means were baseline 0.171038, free prose 0.146103, two-step field 0.103278, and same-turn field 0.098425. A later placebo-control test weakened the claim: among the 30 rows with complete structured outputs, baseline mean Brier was 0.078000, two-call prose 0.107254, same-turn field 0.110300, free prose 0.122767, and two-step field 0.149921; ten additional Codex rows failed at runtime before emitting forecasts. This is not an established mechanism. It leaves structured evidence fields as a hypothesis for larger paired tests, but argues against treating the stronger two-step break as established.

Bias-transfer diagnostic. We also tested whether human-style cognitive-bias categories predicted LLM forecast distortions. The useful distinction is representational: some biases have a large natural-language footprint, some depend on human utility or affect, and some are heavily discussed in case-study text even though the model has no corresponding utility function. This distinction helped organize prompt-risk and family-sensitivity tests, but it did not establish a forecasting intervention. In the 180-call anti-bias-prompt test, the average collapse pattern was underpowered, the class-label shuffle was null ($p = 0.5387$), and the raw-gap-adjusted coefficient for text-discussed motivational cases was negative (-0.076587 , $p = 0.0025$). A matched audit using the stored rows also found poor support overlap: at raw-gap caliper 0.05, within-family matching left only 16 with-replacement / 15 no-replacement pairs and flipped the estimated collapse effect negative (-0.072750 , $p = 0.0008$; greedy no-replacement -0.077561 , $p = 0.0006$), while the no-caliper positive estimate relied on large raw-gap distances. The supported claim is therefore diagnostic: bias-category measurements help explain prompt fragility, but a causal cognitive-debiasing intervention requires new matched strata or randomization.

Family/post-training overlay. Family identity and post-training choices reshape per-family channel surfaces, but this paper observes only cross-family contrasts and does not identify the causal role of RLHF. Family-level worry/error sign differences (§5), per-family channel-decomposition R^2 differences (§5), and the self-assessed channel-choice failure where three of five families pick the worse channel at $p \leq 0.005$ each (§5) all live on this axis. The empirical signature is that family

identity, conditional on channel and contract, is itself a central predictor.

The methodological backbone is orthogonal to the three axes. A power-aware scoring discipline (Fisher- z power calculation before launch, three possible outcomes after scoring, equivalence testing, BH-FDR, LOO- R^2 at small N) sits underneath every axis-1/2/3 measurement. The discipline produced the study’s most consequential retraction: 8 of 12 prior “no effect” claims became underpowered rather than negative findings, and the corpus-validity-drift filter (`resolve_date > max(panel_cutoff)`) empties the most-cited 2024 LLM-forecasting benchmark for the current LLM generation (§10).

5 Elicitation diagnostics: channels are conditional, not decision rules

The uncertainty-channel experiments explain why raw probabilities alone are insufficient, but they are not themselves decision rules. We elicited worry scalars, bid-ask spreads, trajectory variance, frequency framings, self-predicted Brier intervals, outside-view base rates, and self-assessments about which channel should predict error. The pattern is consistent across these probes: LLMs emit useful side information about contracts, families, and sources, but the side information is conditional. It changes sign by family, weakens under task-difficulty controls, and often fails when moved from a low-overlap corpus to public-domain market questions.

The channel material has a narrow role in the argument. The diagnostics support one claim needed for the harnessing thesis: raw LLM forecasts contain more structure than a single probability, and that structure requires label-time-valid scoring, source and family splits, and comparison against simple baselines before use.

Several negative results are central. Channel-only decision rules that look attractive in small or low-overlap cells fail under larger leave-one-out checks or source controls. Models can identify contracts that feel leaky or risky without translating that feeling into better probabilities. Self-assessed channel selection can even be anti-coherent: in the low-overlap channel-choice probe, three of five families often selected the channel that was worst for their own error. These failures motivate the paper’s emphasis on source-valid calibration, pairwise ranking, source-specific checks, and future allocation tests rather than generic self-monitoring or prompt-only intervention.

6 Additional all-channel diagnostics

The all-channel prompts collected several side measurements in a single forecast pass or as paired conditions. Five additional findings emerge at $N=131\text{--}142$ public-domain:

Rollback hurts. A self-counterfactual prompt (“imagine you were trained only up to 2023, emit a rollback forecast”) worsens Brier for 4 of 5 families at $N=142$ paired (Δ Brier rollback–normal: claude +0.064, codex-5.5 +0.034, codex-5.4-mini +0.028, deepseek +0.028, gemini −0.005). Post-2023 training is net-useful even though families cannot introspect which contracts benefit. *Practical caution:* avoid “imagine you knew less” framings on forecasting tasks without a heldout correction check.

Leakage agreement without calibration. On the same rollback prompt, all 10 pairwise $\rho(|\text{leak}_A|, |\text{leak}_B|)$ between families are $\geq +0.43$ (claude $\leftrightarrow$ codex-5.5 $\rho = +0.80$). Families agree on which contracts feel high-leakage. Per-family $\rho(|\text{leak}|, \text{err}^2)$ is nevertheless equivalent to zero for all five at the ± 0.30 target. Self-reported leakage is a contract-structural signal, not a per-family calibration channel. It is useful as a corpus-curation filter for low-leakage LLM-fresh tests.

Diagnostic	Main evidence	Manuscript role
Worry scalar	Pool-level worry is a positive tail-risk signal at $N=590$, but per-family worry-Brier signs split. Claude and GPT-5.5 can be most worried when most right, while GPT-5.4-mini is direction-sensible. Topic-trigger and mean-regression probes explain the sign flip better than rationale length, wallclock time, or hedging vocabulary.	Use worry as a behavioral channel only after conditioning on family, corpus, and source. It is not a universal calibration transform.
Cognitive text versus calibration	Failure-mode words in rationales are weakly positive for the original trio but near zero for Gemini and DeepSeek at roughly $N=60$ per family. Stake framing raises worry for every family while leaving point probabilities nearly unchanged.	The model can produce diagnostic language without recomputing the forecast from that language. This supports interface discipline, not free-form chain-of-thought reliance.
Bid-ask and frequency probes	Bid-ask spread is positive on the low-overlap corpus for GPT-5.4-mini and DeepSeek, but public-domain replication is underpowered or sign-flipped. Frequency framing improves GPT-5.4-mini on low-overlap questions and does not transfer to the public corpus.	These are promising elicitation probes. They do not justify a source-general policy.
Orthogonality and multi-channel fits	Worry, spread, and trajectory variance are nearly uncorrelated at expression level. Early same-forecast R^2 survives leave-one-out only for GPT-5.5 on the $N=42$ public corpus; the larger five-family audit gives negative leave-one-out R^2 for channel-only Brier prediction across all five families.	Channels are distinct measurements, but distinct does not mean ready for use. They are useful for diagnosis and design of later scoring rules.
Conditional structure and family allocation	Sub-source decomposition flips channel signs across yfinance, Manifold, and Polymarket cells. Pooled channel value collapses after task-difficulty controls. Easy-contract cells show headroom, hard cells overfit, and naive mean/median panels do not beat the best single family.	Forecast signal lives in family-by-source-by-contract interactions. Current observable allocation rules find headroom but do not yet recover it under controls.

Table 3: Uncertainty-channel experiments as diagnostics. The shared lesson is conditionality: emitted channels can locate signal, but they have not become source-general probability rules.

DeepSeek’s emitted decision threshold is calibrated. In an asymmetric-loss decision-threshold prompt (commit-YES threshold under symmetric / YES-bad-3 $\times$ / NO-bad-3 $\times$ loss), Codex-5.4-mini emits the most Bayes-coherent thresholds (mean YES-bad 0.750, exact match). For DeepSeek, $\rho(\text{threshold_yes_bad}, \text{err}^2) = -0.19$, 95% CI $[-0.35, -0.03]$ at $N=142$. The emitted threshold tracks accuracy for one family. This is a novel diagnostic channel; policy use requires a heldout utility test.

Stake-framing amplifies worry universally, leaves p unchanged. Same contract framed as “\$1 stake” vs “\$100K stake” across 103–142 paired contracts per family. Worry-scalar amplifies for every family (Δworry from +10.8 deepseek to +33.5 claude on the 1–100 scale). But $|\Delta p_{\text{success}}| < 0.03$ for every family. This cognitive-shift-without-calibration-shift result connects the cognitive-decoupling result to the stake-framing and risk-amplitude literature in human decision-making [6].

Self-assessed channel choice fails for three of five families. A channel-choice prompt (“which of your own uncertainty channels best predicts your error?”) reveals systematic anti-coherence at $N=15$ on the low-overlap corpus: claude, codex-5.5, and deepseek pick the worse channel 87% of the time (binomial $p=0.005$ each). Within those picks, $\rho(\text{picked}, \text{err}^2)$ is negative while $\rho(\text{unpicked}, \text{err}^2)$ is strongly positive (claude +0.51, codex-5.5 +0.53, deepseek +0.65). The candidate inversion rule is: when these three families claim “worry is my best channel,” test spread instead (or the analogous flip). Codex-5.4-mini is coherent (73% right); gemini is random. Companion finding to the per-family worry sign-flip, on a new axis. This result is low-overlap-only at $N=15$; cross-corpus replication at $N=42$ external is pre-registered. The pattern (LLMs have contract-level structural intuitions without calibration to truth) recurs across the rollback, leakage, decision-threshold, and channel-choice results, the defining negative finding about LLM self-monitoring.

7 Applied calibration and allocation consequences

The channel and family diagnostics become useful only if they change a scored forecast or a decision rule. This section separates three evidence grades: robust enough for a source-valid calibration view, useful as a design clue, and unsupported as an applied rule.

Pattern	Evidence	Status
Low-probability overconfidence	In the lowest forecast quintile, every family underpredicts YES outcomes; gaps range from +0.34 to +0.72.	Basis for the low-probability calibration rule, subject to source-currency checks.
Horizon and source difficulty	Longer horizons have higher Brier ($\rho = +0.161$, 95% CI $[+0.026, +0.290]$); Polymarket is harder than Manifold, which is harder than yfinance for every family.	Design clue; not enough alone for an applied rule.
YES underprediction	On contracts resolving YES ($N = 26/42$), every family’s mean p_{success} is below 0.5.	Supports calibration diagnostics.
Panel agreement and naive ensembles	Cross-family forecast correlations are high (mean $\rho = +0.72$); mean/median-of-five have higher Brier than the best single family on the $N = 142$ paired set.	Warns against naive averaging.

Table 4: Cross-family regularities that motivated calibration and allocation tests. They are useful only when converted into a scored rule and compared with simple baselines.

The composed four-rule recipe converted these patterns into a scored forecast by applying a low-probability discount, a middle-band YES correction, a horizon shrinkage, and a source-difficulty shrinkage to the mean panel forecast. The recipe beats naive aggregation on the same $N = 142$ paired set, but it does not clear the stronger best-single and source-balanced policy bars.

strategy	Brier	Δ vs. best-single	90% CI	p_{perm}
median-of-5	0.272	+0.018	–	–
mean-of-5	0.261	+0.007	–	–
best-single (Claude)	0.254	–	–	–
adjusted aggregate (universal-only)	0.232	–0.022	[–0.050, +0.005]	0.18
adjusted aggregate (per-channel)	0.246	–0.009	[–0.035, +0.016]	0.58

Table 5: Composed adjustment on the $N = 142$ paired public-domain set. The within-cohort win is against naive aggregation, not against the strongest simple correction.

Three points matter. First, the adjusted aggregate improves over naive aggregation within this cohort: it beats median-of-five by -0.040 Brier ($p = 0.0013$) and mean-of-five by -0.029 ($p = 0.0069$). Second, the comparison with the best single family remains underpowered: the point estimate is favorable (-0.022) but the paired test is $p = 0.18$, with an estimated $N \approx 250\text{--}300$ required for the observed effect. Third, later source-balanced audits weaken the composite. The current source+ σ allocation rule and diagnostic-triggered allocation have higher Brier than simpler rules; Hedge over raw-family, low-probability-adjusted-family, and simple-pool experts is directionally better but nonsignificant (0.226481 versus 0.233529, $p = 0.4671$) and regresses on Manifold in the balanced slice. Choosing the best family in hindsight remains far better (0.117454 Brier), so family-choice headroom exists, but the observable rules have not recovered it.

The pairwise ranking packet supports a controlled relative-judgment use. A source-balanced four-family contrastive packet supports pairwise ranking over 24 unique non-tie pairs (accuracy 0.750, utility +0.583, $p = 0.0044$ versus random). Translation tests are favorable but not yet ready as a point-probability layer: same-packet translated-minus-low-probability-correction is -0.022714 with $p = 0.0628$; one cross-packet direction clears $p = 0.0314$, the reverse direction misses at $p = 0.0636$; and the joined market-control slice is only 24 rows with translated-vs-market $p = 0.5783$. A prospective Polymarket packet has frozen 24 pairwise market bars before model calls, but unresolved outcomes cannot support a current scoring claim. Pairwise ranking therefore appears in the paper as a controlled use, while the current point-probability rule remains the low-probability calibration rule.

The low-probability correction as a standalone rule beats raw on every family at $p < 0.05$. The rule applies only when the panel forecast is below 0.20, shrinking that low forecast toward 0.10 by $\hat{p}_{\text{low}} = 0.35\bar{p} + 0.65(0.10)$. The simpler form of the recipe, the single low-probability discount applied per-family without the other three rules, improves per-family Brier on every panel member at $p < 0.05$:

family	raw Brier	discounted Brier	Δ	paired p_{perm}
claude	0.2543	0.2240	–0.0302	0.016
codex-5.5	0.2625	0.2416	–0.0208	0.030
codex-5.4-mini	0.2714	0.2450	–0.0264	0.015
deepseek	0.3222	0.2704	–0.0518	0.0008
gemini	0.3167	0.2840	–0.0327	0.008

Discounted Claude at Brier 0.2240 is at parity or better than the four-rule adjusted aggregate

(0.232), so the additional features add little on this corpus. The single low-probability adjustment improves Brier at $p < 0.05$ across every model family, including DeepSeek and Gemini, which were not the panel that originated the rule. A fitted-calibrator audit using the stored rows did not identify a better replacement: source-isotonic slightly improved the overall point estimate but lost Manifold and was nonsignificant, while tail-beta shrinkage was worse. The 2026-06-04 source-documented rerun excludes 10 yfinance/yfinance_etf complete panels lacking label-time documentation, leaving 132 panels; source-isotonic remains unsupported at -0.005248 Brier versus the low-probability correction with paired $p = 0.7099$, while raw mean-panel is $+0.029598$ worse. A later source-currency stress audit narrows the rule: it improves post-cutoff rows (-0.025326 , tail-only -0.101306) and regresses pre-cutoff/source-visible rows ($+0.035016$, $p = 0.0002$; tail-only $+0.097719$, $p = 0.0002$). A row-level rerun through the shared source-currency discriminator leaves those scores unchanged while exposing 39/240 stored-flag-vs-computed-relation conflicts. The low-probability rule is therefore forward-looking calibration tied to computed cutoff and label-time documentation, not retrospective benchmark correction.

8 Harnessing LLM forecast signal under constraints

The equal-information market controls change the paper’s question. The defensible unit is a validated and controlled forecast row: the completed Polymarket slice clearly favors the market, and the Manifold second-source slice favors the market but is inconclusive. The useful finding is that the market result does not make the LLM signal disappear. It changes which part of the signal can be used. In the current evidence, the usable pieces are a source-valid calibration rule, a pairwise ranking task, and measurable family-choice headroom. These mechanisms belong in the same paper because they depend on the same measurement foundation that makes the market controls interpretable.

The low-probability rule is the current point-probability use case. The low-probability discount is the strongest practical rule in the database. It improves every tested family on the public-domain panel, including families that were not used to originate the rule. On source-documented rows the raw mean-panel remains worse than the low-probability correction by $+0.029598$ Brier with paired $p = 0.0062$. On the source-currency stress panel, the same rule improves post-cutoff rows by -0.025326 while regressing pre-cutoff/source-visible rows by $+0.035016$ with $p = 0.0002$. This is the pattern in miniature: the rule extracts signal from the forward-looking slice and damages retrospective/source-visible rows. It is therefore a source-valid calibration view, not a universal correction.

Pairwise ranking is the current relative-judgment use case. The strongest positive ranking evidence supports relative judgment rather than absolute probability. A source-balanced same-source/minimal-pair packet gives 24 unique non-tie pairs with accuracy 0.750, utility $+0.583$, $p = 0.0044$ versus random, and $p = 0.0002$ versus source control. Later probability-translation tests are favorable in one direction, while the required single-contract probability checks remain unmet: same-packet translated-vs-low-probability-corrected and translated-vs-raw checks, bidirectional cross-packet transfer, joined market control, and prospective causal-order resolution are not all satisfied. The supported use is pairwise ranking or tournament support.

Interface and family-choice evidence shows where the next gain could come from. Structured evidence fields are plausible because some small tests favor typed fields over free prose, but the placebo-control continuation is negative for the stronger two-step claim. Family choice has real headroom: choosing the best family in hindsight reaches much lower Brier than current decision rules, and family-by-contract interaction is substantial. Current observable selection rules do not recover that headroom: Hedge over raw, low-probability-corrected, and simple-pool experts

is directionally better but not significant, and graph-family weighting is small and fragile. These results locate signal; they do not yet supply a reliable family-selection rule.

Prompt-only improvement survives only as failed evidence plus a narrower interface hypothesis. Generic reflective prompting, selective action, and self-repair do not reliably improve forecasts under controls. The tested prompt-intervention variants either fail confirmation, overcorrect, produce no measurable change, or underperform low-probability calibration. This does not rule out tool-using, retrieval-grounded, expert-written, or heldout-tuned prompting systems. It does rule out the simple claim that asking the model to reason harder, repair itself, or act selectively is enough to improve forecast probabilities.

The manuscript therefore stays integrated for now. The positive mechanisms depend on the same validity checks that make the market controls interpretable. A later split is justified only if prospective ranking, calibration, and family-choice evidence becomes large enough to support an independent mechanisms paper.

9 Secondary diagnostics: bias transfer and prompt stability

The bias-transfer and prompt-stability experiments are not the paper’s central contribution. They appear in the main text as diagnostics for why naive prompt interventions and family-general allocation should not be trusted.

Bias transfer. Three novel-bias tests on the same $N = 30$ public-domain contracts across five families found a structured split: loss-frame and probability-weighting probes were close to symmetric or near-linear, while current-state/status-quo framing moved several families substantially. A held-out seven-bias slate later supported the representational distinction between text-discussed motivational cases and direct utility-like mechanisms. The anti-bias-prompt companion test was weakened by raw-gap controls, and the out-of-distribution slate was inconclusive and confounded by cross-family pretraining differences. We therefore use these results only to interpret family/source sensitivity and prompt-risk, not to claim that cognitive-debiasing prompts improve forecasting skill.

Prompt stability. The prompt-invariance audits give the same warning from another angle. Standard forecast prompts yield relatively stable point estimates across all five families, but reference-class outside-view prompting is a different reasoning architecture rather than a harmless paraphrase. The worry channel is much noisier across prompts than p_{success} , and cross-prompt worry-Brier signs can reverse relative to the low-overlap experiment. Any decision rule that depends on a verbal uncertainty channel must therefore be treated as family-, corpus-, and prompt-bound until it clears heldout controls.

These diagnostics bound the harnessing thesis: model signal does not become reliable through generic reflection, debiasing, or self-repair. It becomes usable only when the interface is tied to a scored channel, a source-valid calibration rule, a pairwise ranking task, or a predeclared allocation rule that beats simpler controls.

10 The re-audit discipline: a warning to the field

The discipline that produced the results above also produced a re-audit of the study’s own prior findings. We treat the re-audit itself as a contribution because the same overcalling pattern almost certainly affects the published LLM-forecasting literature at the sample sizes it currently reports.

10.1 Power calculus applied to our own study

Before applying the power-aware audit, eight prior “no effect” or “cross-corpus replication” claims in the study had been written down as findings. Applying the Fisher- z audit retrospectively:

- Four “null” findings move from ruled out to underpowered: the data did not actively rule out a meaningful effect at the per-family target $|\rho|$ floor; the absence-of-signal was an absence-of-power.
- Five cross-corpus claims move from “replicated” or “failed-to-replicate” to corpus-specific: the original cross-corpus reading was a single sub-source point estimate at $N=5-15$ external, with CIs that crossed zero and with sub-source heterogeneity unmodeled.
- One self-knowledge-inversion finding (originally three families at $N=5$) retracts to a single family at $N=15$ on the low-overlap corpus; the other two cross-corpus directions flip on rerun.
- Two per-family decision rules (Brier- Δ map and *consider-both-sides* effect) move from usable to “directional hypothesis only”; the required N to confirm a $\Delta_{\text{Brier}}=0.05$ effect at 80% power is ≈ 459 per family, an order of magnitude above the original $N\approx 30$ per cell.

In aggregate, $\approx 50\%$ of the study’s prior conclusions changed under the power-aware re-audit. None of the underlying executions were flawed. The interpretation discipline was.

10.2 A worked example: the per-family worry sign-flip retest

The previous Gemini worry-channel hypothesis was scope-restricted to “low-overlap corpus, original prompt only” pending a ~ 42 -call retest of the same prompt on the public-domain $N=42$ slice used by the other public-domain tests. The retest, run with the same worry-only prompt on `gemini-2.5-flash` via API:

- Schema-OK: 42/42.
- $\rho(\text{worry}, \text{Brier}) = +0.110$ – the direction-correct direction, not the inverted direction the prior rule had assumed.
- Fisher- z detectable $|\rho|$ at $N=42$, 80% power: ≥ 0.43 . Observed $|\rho|=0.11$ is well below this; the CI excludes the original “inverted” direction ($\rho < -0.30$).
- The inversion hypothesis is ruled out at the target effect size; the direction-correct point estimate remains underpowered for the positive hypothesis.

The Gemini channel rule stays withdrawn. The earlier “inversion” claim, estimated from $N=5$ with $N_{\text{wrong}}=2$, was a sample-size effect. This is the lowest-cost decisive retest available; two comparable retests remain open.

10.3 Implications for the published LLM-forecasting literature

Two prominent 2024 contributions to LLM forecast calibration are (a) the ensemble-prediction result that a 12-LLM aggregate matches human-crowd Brier on real binary questions [14] and (b) the retrieval-augmented LLM that approaches human-superforecaster accuracy on a contamination-resistant slice [5]. Both report claims at sample sizes that, under the same Fisher- z power calculus we apply to our own study, are limited by:

- For a Brier-delta $\Delta=0.05$ comparison (LLM ensemble vs. human crowd) at 80% power, paired permutation requires $N \geq 300$ –500 per side.
- For a per-model rank correlation $|\rho|=0.30$ at the same power: $N \geq 91$.
- For a per-(model, sub-source) cell at $|\rho|=0.40$: $N \geq 50$ per cell – requiring sub-source $N \geq 250$ across five families when sub-sources are five-way mixed.

We do not re-analyse those datasets here; we observe only that the standard “2024-style” aggregate claim is structurally compatible with the same overcalling failure mode our own study exhibited. Paleka et al. [11] make the broader version of the same warning: LLM forecasting evaluation is unusually vulnerable to temporal leakage, unreliable date-restricted retrieval, benchmark extrapolation errors, and circular comparisons against human forecasts already visible to the model or retrieval system. Consistency-check benchmarks offer a complementary way to probe coherence before outcomes resolve [12], but they do not remove the need for resolved, same-information scoring. The methodological gap this paper targets is a consistent power-aware audit: every claim labelled with one of three possible outcomes, n_{required} pre-computed before launch, sub-source heterogeneity reported alongside the aggregate, and prior verdicts re-audited under the same calculus as new ones.

The external bar has also moved since those papers. The recent literature now has four distinct benchmark classes.

- **Future-question benchmarks.** ForecastBench makes future-only benchmark generation explicit and reports expert forecasters outperforming the top LLM on its initial human/LLM sample [8]. AIA Forecaster reports human-superforecaster-level performance on ForecastBench and additive value when ensembled with market consensus, while also finding that the system underperforms market consensus alone on a harder liquid-market benchmark [1].
- **Belief-updating benchmarks.** EVOLVECAST tests whether LLM forecasts move appropriately when new post-cutoff information is supplied, and reports partial responsiveness but inconsistent or overly conservative updates relative to human references [15]. That result is complementary to our label-time and source-currency checks: both settings show that a forecast row is not defined only by the final event label, but also by the information state available when the probability is produced.
- **Trading and replay benchmarks.** Prediction Arena evaluates agents with real capital on Kalshi and Polymarket and finds platform-specific performance differences rather than a generic forecasting edge [16]. PolyBench records timestamp-locked Polymarket order-book/news snapshots and reports that most models have negative returns despite high stated confidence [3]. PredictionMarketBench emphasizes execution-realistic replay because profit and loss mix probability accuracy with fees, liquidity, position sizing, and settlement risk [2].
- **Market-as-evaluation infrastructure and coordination.** Repetto-style systems position prediction markets as an AI-training and evaluation substrate [13]. Foresight Arena separates proper probabilistic scoring from trading profit by using commit-reveal forecasts and an alpha-over-market score; its power analysis estimates that detecting a small edge over market consensus requires hundreds of resolved binary predictions [10]. MarketBench studies a related coordination problem: agents are miscalibrated about their own success probability and cost, and capability information improves that calibration only modestly [4]. That result aligns with our family-choice findings: conditional headroom can exist even when current observable selection rules do not safely recover it.

Read against that literature, the present paper is not a live-market benchmark or an autonomous-trading evaluation. Its distinct contribution is the validity layer those benchmarks now make unavoidable: before comparing an LLM, a market, and a human, the row must specify what was source-visible, what label vintage is admissible, and whether the baseline was measured under equal information. This is also why evidence that markets score better is scientifically useful. If an LLM has higher Brier after the information state is equalized, the remaining question is whether any model-derived signals remain usable under calibration, ranking, and family/source constraints.

What a field-wide audit would measure. The evidence in this paper is enough to show that row-level validity can change conclusions in this program. It is not yet enough to claim that the same failure rate holds across the field. A field-wide audit would treat each benchmark row as the unit of analysis and record the following checks before comparing model, human, or market scores:

Audit check	What must be recorded	Why it matters
Source currency	Forecast timestamp, model cutoff or retrieval window, resolution date, and whether the resolved answer was source-visible at generation time.	Separates future-event prediction from retrieval or source familiarity.
Label-time validity	Outcome label, label source, data vintage, and any later revisions or settlement-rule changes.	Prevents scoring against values that were not admissible at resolution time.
Equal-information comparator	Human, crowd, market, or agent baseline measured on the same contract at the same pre-outcome information time.	Prevents comparing a model forecast with a comparator that knew more or less.
Effective sample size	Event-family identifiers, source strata, repeated sibling markets, and family/model repetition.	Prevents row-rich but event-thin conclusions.
Decision rule status	Whether the tested rule was predeclared, tuned on the same rows, or evaluated prospectively.	Separates exploratory diagnostics from supported use.

Table 6: Field-wide validity-audit protocol. The present manuscript supplies this protocol and a scored within-program audit; a cross-benchmark failure-rate claim requires applying the protocol to several public benchmark families.

For reproducibility, `field_wide_validity_audit_protocol.py` materializes this protocol as a row schema and seed matrix over ten benchmark families: ForecastBench, the Halawi 2024 binary-resolved benchmark, AIA Forecaster, Prediction Arena, PolyBench, PredictionMarketBench, Foresight Arena, EVOLVECAST, MarketBench, and Reppo-style market infrastructure. A companion script, `field_wide_validity_local_evidence.py`, records the local Halawi date-distribution summary with its limitation: the raw benchmark rows are not present locally, so this is a warning about a likely replication filter, not a completed external row audit. The protocol output is therefore a continuation plan, not field-wide evidence.

10.4 Source-currency, label-time, and equal-information audit

The central re-audit result is that benchmark validity is a moving target. A dataset that was a fair future-event benchmark for one model generation can become a retrieval or source-familiarity

benchmark for a later generation. The local claim register records a Halawi 2024 binary-resolved date histogram with no 2025-or-later resolutions; for current model generations, the corresponding eligibility filter, $resolve_date > \max(panel_cutoff)$, would leave zero rows in that local summary. This is not a replacement for a raw-row external audit, but it is the concrete corpus-validity drift warning that motivates the field-wide protocol.

The source-currency panel is matched on source, topic, question-length bucket, and cutoff relation. Reliable base-rate bands are not available for that panel, so base-rate matching remains a stated limitation rather than a hidden control. The database now contains 103 external market baseline rows: 51 not-equal-information Manifold diagnostic rows, 4 equal-information Polymarket rows from the failed seven-day packet, 24 equal-information Polymarket rows from the replacement packet, and 24 equal-information Manifold rows from the 2026-06-15 history fill. The two completed 24-row equal-information slices are the relevant market controls for this manuscript. They do not support raw LLM superiority: Polymarket is decisively better, and Manifold is directionally better but underpowered.

This audit changes what the paper can support. The Manifold source-currency result is strong enough to report as a measurement-validity contribution. The FRED work shows why label-time repair is not optional for official-data questions. The equal-information market rows rule out a broad market/human superiority claim while sharpening the harnessing thesis: useful LLM signal has to be extracted through source-specific calibration, pairwise ranking, structured interfaces, or future allocation rules that beat the market/human bar under the same information rule.

10.5 The applied consequence

The re-audit rule: *a per-family or per-(family, sub-source) decision rule should be treated as supported only when its supporting CI has been computed on $N \geq n_{\text{required}}$ at the target effect size, and only with sub-source stratification reported.* Three remaining low-cost retests (~ 0 , 42, and 420 calls respectively) close the same gap on the study’s other decision-rule hypotheses; their results are published alongside the existing rules as the retests complete.

11 What this paper does not establish

- It does not show that LLMs beat humans, human crowds, or prediction markets. The database has 103 external market baseline rows, including 52 equal-information market rows across Polymarket and Manifold. On the completed 24-contract Claude+Codex+Gemini+DeepSeek Polymarket replacement slice, the market baseline beats the four-family model panel. On the separate 24-contract Manifold fill, the market is also ahead of the selected five-family low-stake model panel, but the paired comparison is inconclusive. Broad superiority would require predeclared or sufficiently powered source-balanced baselines under the same pre-outcome information rule.
- It does not establish translated pairwise probabilities as a standalone probability layer. Pairwise ranking survives as source-heldout evidence with promising translation tests, but same-packet, cross-packet, market-control, and prospective causal-order checks remain incomplete.
- It does not establish a source-general source-currency result beyond the main Manifold panel. Polymarket tests are aggregate-positive but matched-stratum null/opposite-sign, and FRED current-label positives weaken under label-time/vintage repair.
- It does not show that cognitive-debiasing interventions never transfer to LLMs. Four of five tested effects are underpowered at our low-overlap-corpus N , not falsified.

Audit lane	Current evidence	Conclusion
Manifold source-currency panel	80 matched contracts / 240 tool-free calls; post-minus-pre Brier +0.191098, paired-stratum delta +0.2155, permutation $p = 0.0004$. Partial market-prior repair joins 51 contracts and leaves the effect positive under adversarial missing-band sensitivity.	Strong candidate measurement result: source-visible and post-cutoff rows cannot be pooled without documentation.
Manifold pre-outcome market bar	On the 51 repaired contracts, market Brier is 0.099673 versus joined LLM-panel 0.166963 overall; market+LLM blending is unsupported ($p = 0.794$). These rows are typed as not-equal-information external baselines.	Useful diagnostic comparison, but not a broad human/crowd baseline.
Polymarket source check	Gemini and DeepSeek aggregate post-minus-pre directions are positive, but matched source/topic/length strata are null or opposite-sign. The initial seven-day market-price freeze design only fills 4/24 rows; horizon sweep tops out at 12/24.	Diagnostic evidence, not a source-general replication or usable equal-information design.
Polymarket replacement equal-information packet	Replacement sampling on 2026-06-15 selects 24 one-per-event rows from 80 eligible candidates at a two-day freeze horizon, 14 NO / 10 YES, all open-by-target with nonempty CLOB history. The four-family panel has higher Brier than the market: panel 0.267758 vs market 0.072964, paired $p = 0.0068$.	Clear same-information market advantage.
Manifold equal-information packet	Public API history fill validates 24/24 rows, 15 NO / 9 YES. Selected five-family panel Brier is 0.198723 vs Manifold 0.160977, panel-minus-market +0.037746, paired $p = 0.5431$.	Second-source comparison; market advantage is directional and inconclusive.
FRED / official-data lane	Fixed one-year FRED companion supplies 49 pre-cutoff rows; full paired post-minus-pre delta is only +0.016477 ($p = 0.30375$). Blinded-control apparent current-label penalty collapses from +0.024719 to -0.002989 after vintage repair.	Official-data diagnostic lane; label-time checks are required before current-label positives count.
Metaculus target cells	Current authenticated endpoints expose post/question payloads but not resolved binary values plus dated aggregate history for the sampled rows; data-download endpoint is restricted.	Remains a data-access question, not a negative result.

Table 7: Re-audit evidence by lane. The paper’s broad comparison claims depend on source currency, label-time validity, and equal-information status, not merely on model-call volume.

- It does not establish that GPT-5.4-mini’s low-overlap-corpus frequency-framing improvement would survive cross-corpus replication at proper power. $N \geq 91$ public-domain is required.
- It does not generalize beyond five model families and two corpus classes. The low-overlap corpus is private and not publicly available in raw form.
- It does not decompose LLM *reasoning* mechanistically. The channel-orthogonality claim (§5) is at the *expression* level; a reasoning-decomposition claim would need white-box mechanistic interpretability.
- The four-axis confounding between corpora means the cross-corpus result is evidence that at least one axis matters, not yet a clean attribution. The 4-cell de-confounded corpus is the most important methodological follow-up.
- It does not establish that RLHF/alignment *causes* the per-family bias differences. Every result in that group, in-distribution and out-of-distribution, is a *cross-family* contrast, which confounds alignment with pretraining-corpus differences; families differ in both. The apparent “more alignment, less bias transfer” ordering also runs opposite to the established result that instruction-tuning amplifies several cognitive biases [7]. The supported claim is therefore observational—*family identity predicts bias magnitude*—not a causal post-training explanation. Isolating the alignment stage requires within-family checkpoints (pre- vs post-RLHF) or matched-pretraining families, which we do not have.
- Bid-ask spread cross-corpus is currently underpowered; the direction-shift is suggestive but not conclusive at $N=42$.
- The worry-Brier sign-flip mechanism is provisional (§5 multi-probe synthesis); activation-level probing and a topic-matched corpus would settle it.
- Per-family signal strength is conditional on prompt-invariance (§9). Gemini and DeepSeek findings should be read with the prompt-stability discount documented there until the within-family paraphrase test is run.

Next tests. Table 8 states the evidence required to strengthen each claim. The most direct route to a broader measurement contribution is a field-wide validity audit of public forecasting benchmarks and market-replay environments. That route does not require LLMs to beat markets; it requires showing that row-level source-currency, label-time, or equal-information checks are often missing and can change conclusions. A broad LLM-vs-market or LLM-vs-human claim requires a predeclared or substantially larger source-balanced equal-information packet that beats the market/human baseline, not more calls on the current rows. A source-general source-currency claim requires Metaculus/export access or another non-Manifold panel with matched pre/post resolution-date coverage and label-time documentation. Pairwise ranking becomes a standalone probability layer only if same-packet, cross-packet, market-control, and prospective causal-order checks all clear. The main route to a stronger intervention paper is a public structured-metacognition experiment: bare prompt, length-matched placebo, expert-training prompt, audit-informed prompt, and failure-mode-specific prompt on the same source-valid external corpus. The low-overlap-corpus findings become externally general only after a sanitized release or a public niche-domain replication. A companion evidence matrix records each candidate result, the checks it has passed, the checks still missing, and the next decisive test. If those tests fail or remain underpowered, the paper remains what the present evidence supports: a measurement-validity contribution with controlled calibration and ranking consequences.

Claim to strengthen	Evidence required	Current status
Field-wide benchmark validity	Row-level audit of public forecasting benchmarks and market-replay environments, recording source currency, label-time validity, equal-information status, and conclusion changes after repair.	Not claimed here; current evidence shows the failure mode inside this program, not field-wide prevalence.
LLM vs. market/human performance	Predeclared, source-balanced equal-information packet with enough resolved rows to beat the market or human baseline under proper scoring.	Not supported; current equal-information slices favor markets or are inconclusive.
Source-general source-currency result	Non-Manifold panel with matched pre/post rows, admissible label vintage, and base-rate documentation.	Supported on the main Manifold panel; other sources remain diagnostic.
Pairwise ranking as probability layer	Same-packet and cross-packet ranking replication plus prospective probability translation against raw, calibrated, and market controls.	Ranking use supported; probability translation remains provisional.
Structured metacognition intervention	Pre-reviewed public-corpus experiment comparing bare prompt, length-matched placebo, expert-training prompt, audit-informed prompt, and failure-mode-specific prompt under source-valid scoring.	Designed as the strongest continuation test; the partial 131/600 Gemini run does not yet support the intervention claim.
Structured evidence fields	Larger balanced paired test that beats free prose, same-turn fields, and two-call prose after runtime failures are included.	Hypothesis only.
Family allocation	Predeclared observable features or independent reviewer source that recovers best-family headroom after cost.	Headroom exists; current allocation rules do not recover it.
External generality of low-overlap results	Sanitized release or public niche-domain replication that breaks the current novelty/source/topic/horizon confound.	Not yet established.

Table 8: Next tests. These are evidence requirements, not additional claims made by the present manuscript.

12 Conclusion

The main lesson is measurement-first. LLM forecasting evaluations are not interpretable merely because the model output is a probability and the event later resolves. The row must specify what was source-visible to the model generation, whether the label uses an admissible time vintage, and whether the human or market comparison was measured under the same information rule. In this study, those checks change the claim: the Manifold source-currency panel supports a measurement-validity result, while the same-information Polymarket and Manifold market comparisons rule out a raw LLM-superiority reading of the current evidence.

The constructive claim is substantive because it identifies where model signal survives strong controls. The usable unit is not the raw panel probability. The current evidence supports source-valid low-probability calibration, pairwise ranking as a relative-judgment task, structured evidence fields as a hypothesis, and family-choice headroom as a target for future selection rules. It also shows that generic reflection, self-repair, and simple prompt intervention are not enough. The paper’s contribution is therefore a validity layer plus a map of which model-derived signals remain worth testing once invalid comparisons are removed.

Future work should not try to rescue the broad claim with more calls on the same rows. The decisive tests are a field-wide row-validity audit, prospective or larger source-balanced equal-information packets, non-Manifold source-currency panels with label-time documentation, and predeclared mechanisms that beat raw, calibrated, and market controls under the same information state.

13 Reproducibility

All model calls persist as JSONL files and are ingested into the SQLite database at `analytics/public/calibration/forecaster_calibration.db`. Pre-registrations and verdict resolutions are in the same database. The reusable general-purpose statistics module is at `src/ztare/experiment_stats.py` (power calculator, bootstrap CI, paired permutation, Fisher- z Spearman, TOST, BH-FDR, power-aware verdict, BIC Bayes factor, reproducibility hash). Forecasting-specific wrappers are in the per-project `calibration_stats.py`; the public methodology file lists the exact commands for each finding.

Reproducing the scope checks. Three audit scripts define the scope boundary:

- `paper_readiness_exhaustion_audit.py`
- `paper_coherence_audit.py`
- `independent_equal_information_source_audit.py`

Equal-information acquisition is split into deterministic scripts:

- `equal_information_baseline_export_packet.py`
- `equal_information_baseline_result_ingest.py`
- `equal_information_freeze_feasibility_audit.py`
- `equal_information_horizon_sweep.py`
- `equal_information_replacement_sample_acquire.py`

- `equal_information_replacement_dispatch_packet.py`
- `equal_information_replacement_score.py`
- `non_polymarket_equal_information_export_packet.py`: emits the current 24-row Manifold packet for a second equal-information source.
- `non_polymarket_equal_information_result_acquire.py`: fills the packet from public Manifold history.
- `non_polymarket_equal_information_result_ingest.py`: ingests the validated rows.
- `non_polymarket_equal_information_score.py`: scores the joined model-vs-market comparison.
- `field_wide_validity_audit_protocol.py`: emits the row schema and 10-benchmark seed matrix for the field-wide validity-audit route.
- `field_wide_validity_local_evidence.py`
Emits the provenance-limited Halawi date-distribution summary used as a warning, not as a completed field-wide audit.
- `claim_gap_matrix.py`: emits an evidence matrix separating supported results from underpowered results, claims not valid for broad conclusions, and claims requiring external data.

Together these tools let a reader verify the current market-baseline coverage, inspect the replacement packet before forecasts, reproduce the first same-contract market comparison, and see exactly what evidence is still missing for a broad human/crowd claim, field-wide validity claim, or stronger intervention claim.

Reproduction status. Table 9 separates what is reproducible now from what requires a sanitized or substitute release. The private low-overlap corpus is the main unreleased component, but it affects only the secondary low-overlap elicitation findings. The source-currency, label-time, equal-information, market-control, and source-valid calibration claims are represented by public-market, official-data, database, scoring, and audit machinery in the repository.

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Component	Current status	Reproduction role
SQLite evidence database	Present at <code>analytics/public/calibration/forecaster_calibration.db</code> .	Reproduces call counts, score joins, source-currency screens, label-time screens, and market-baseline coverage.
Scoring/audit scripts	Present in the project tool directories and listed above.	Recomputes the paper’s readiness, coherence, equal-information, and label-time checks without new model calls.
Evidence matrix	Present at <code>projects/llm_forecasting_calibration_program/paper_alignment_v1/workspace/claim_gap_matrix_2026_06_16/claim_gap_matrix.csv</code> .	Lists each candidate result, its current evidence, missing checks, what would change the interpretation, and next action.
Field-wide audit protocol and local summary	Present at <code>projects/llm_forecasting_calibration_program/paper_alignment_v1/workspace/field_wide_validity_audit_2026_06_16/field_wide_validity_audit_seed.csv</code> and <code>field_wide_validity_local_evidence_summary.md</code> .	Supplies the row schema and benchmark-family seed list for the field-wide route, plus a limited Halawi date-distribution warning; it is not evidence of field-wide prevalence.
Public-market packets	Present for Polymarket and Manifold equal-information comparisons.	Reproduces the market-control boundary claims.
Raw low-overlap questions	Not publicly releasable in raw form.	Needed only for direct replication of the private low-overlap elicitation findings.
Sanitized or substitute low-overlap corpus	Planned release path: neutralized identifiers plus a public niche-domain substitute with the same four-axis profile.	Enables third-party replication of the frequency-framing, bid-ask, and low-overlap channel findings without exposing private workflow details.

Table 9: Reproduction status. The central source-currency and equal-information claims are reproducible from public project files; the low-overlap channel findings require a sanitized or substitute corpus.

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A Evidence ledger for compressed diagnostics

The main text compresses several experiment families for readability. Table 10 records what was compressed and what insight is retained. The detailed run records remain in the SQLite database, the project workspace, and the public methodology/claim-summary files referenced in §13.

B Coverage audit for omitted or deferred work

The project log contains many more experiments than the main paper reports. The inclusion rule is conservative: a result appears in the main text only if it changes the validity layer, a supported use case, a stated limit, or a continuation test. Table 11 records the main excluded or deferred families so that compression does not hide evidence.

Compressed family	Retained insight	Where it is used
Uncertainty channels	Worry, spread, trajectory variance, self-predicted Brier, and outside-view base rates expose distinct side information, but their sign and value are family/source conditional.	Motivates source/family conditioning and prevents a universal channel-rule claim.
Self-assessed channel choice	Several families misidentify which emitted uncertainty channel predicts their own error.	Supports the negative self-monitoring result and the need for scored interfaces.
Universal calibration regularities	Low-probability overconfidence, horizon/source difficulty, YES underprediction, and high family agreement recur across families.	Supplies the candidate features tested by the low-probability rule and composed adjustment.
Composed adjustment and family-choice headroom	The adjusted aggregate beats mean/median in-cohort but fails stronger source-balanced scrutiny; choosing the best family in hindsight remains much better than observed allocation rules.	Separates real headroom from established allocation evidence.
Pairwise ranking and translation	Pairwise contrastive ranking is stronger than raw probability translation; translation remains promising but is not yet supported as a standalone probability model.	Retains pairwise ranking as a controlled use, not a market/human superiority claim.
Source and label-time audits	Manifold source-currency, FRED vintage repair, failed Polymarket freeze design, replacement Polymarket, and Manifold equal-information fill show that validity checks change conclusions.	Forms the measurement-validity contribution and bounds the constructive claims.
Prompt intervention/self-repair	Generic reflection, self-repair, and diagnostic allocation mostly fail or produce no measurable change under controls.	Turns prompt-only improvement into a narrower interface hypothesis rather than a standalone claim.

Table 10: Evidence-preservation table for compressed diagnostics. Compression changes placement, not the underlying scope boundaries.

Work family	Why it is not a main claim	How the insight is preserved
Prompt-intervention variants	Generic reflection, selective action, self-repair, and diagnostic-triggered allocation do not beat the relevant controls.	Reported as negative evidence for unvalidated prompt-only interventions and as a test for future tool-using or retrieval-grounded systems.
Objective effort / coding-task calibration	This is a sibling problem about effort prediction and hidden-test performance, not the same forecast-row validity question.	Excluded from GP-245 unless it later supplies a forecasting-specific intervention that clears source and market controls.
Proof-audit and workflow-only findings	These improve the research workflow but ask reviewers to switch domains away from event forecasting.	Excluded from the main manuscript; relevant only as provenance for the audit discipline.
Low-overlap elicitation retests	Several channel findings are promising but still corpus-bound or underpowered for source-general claims.	Preserved as diagnostics and continuation tests: replicate on a public niche corpus or sanitized release before generalizing.
Fitted calibrators and allocation rules	Source-isotonic, graph-family weighting, and diagnostic allocation have not beaten simpler controls robustly.	Reported as headroom evidence; applied use waits for source-balanced panels or an external reviewer/market expert.
Prospective market-freeze packets	Frozen market bars exist for some future comparisons, but unresolved outcomes cannot score current claims.	Listed as continuation tests; no standalone probability layer until outcomes resolve and market/raw/low-probability controls are passed.
Deconfounded corpus design	The current low-overlap/public split confounds novelty, source, topic, and horizon.	Treated as the key methodological follow-up for external generality, not as evidence already in hand.

Table 11: Coverage audit for work compressed out of the main text. The table separates omission for irrelevance, omission for weak evidence, and deferral because the right outcome data or control does not yet exist.

C Composed adjustment recipe: closed form

Let $\bar{p}$ be the mean of the five families' point estimates p_{success} on a given contract, h the days-to-resolution-from-cutoff, and $s \in \{\text{polymarket}, \text{manifold}, \text{yfinance}\}$ the source identifier. The composed adjusted forecast $\hat{p}$ is:

$$\begin{aligned}\hat{p}_{\text{low}} &= \begin{cases} 0.35 \bar{p} + 0.65 (0.10) & \text{if } \bar{p} < 0.20 \\ \bar{p} & \text{otherwise} \end{cases} \\ \hat{p}_{\text{YES-bias}} &= \hat{p}_{\text{low}} + \begin{cases} +0.06 & \text{if } 0.30 \leq \hat{p}_{\text{low}} \leq 0.55 \\ 0 & \text{otherwise} \end{cases} \\ \hat{p}_{\text{horizon}} &= 0.5 + (\hat{p}_{\text{YES-bias}} - 0.5) \cdot \frac{1}{1+0.01 h} \\ \hat{p} &= 0.5 + (\hat{p}_{\text{horizon}} - 0.5) \cdot w_s\end{aligned}$$

with $w_s = 0.70$ for polymarket, 0.85 for manifold, 1.00 for yfinance. The four coefficients $(0.10, 0.06, 0.01, w_s)$ are not learned from held-out folds; they are chosen from the regularities reported in §7 (low-probability overconfidence, middle-band YES underprediction, horizon-conditional Brier slope, per-source Brier ordering). The per-channel alternative described in the main text substitutes per-family channel weights for the universal w_s ; its weights are listed in the appendix companion file. The implementation is in the forecasting-calibration workspace and reproduces on the database snapshot dated 2026-05-28.